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Agentic AI-Based Afib/AFL Detection from ECG Signals with Explainable Clinical Report Generation

Abstract

ECG records are often available as scanned images, which cannot be directly analyzed by signal-based AI models. This project proposes an Agentic AI-based ECG analysis system that converts scanned ECG images into digital ECG signals through image processing and waveform extraction. A 1D ECG Transformer then analyzes the signal to identify potential abnormal patterns. An AI agent coordinates ECG quality assessment, explainability, biomedical knowledge retrieval using RAG, and confidence assessment. Finally, these results are combined to generate an explainable ECG report that supports clinical decision-making.

Keywords:

Agentic AI, ECG Digitization, 1D Transformer, Explainable AI, RAG, Biomedical AI, Clinical Decision Support

1. Research Domain

- **Primary Domain:** Artificial Intelligence and Machine Learning
- **Sub-Domain**: Biomedical AI and Explainable Deep Learning
- **Application Area:** ECG Analysis and Clinical Decision Support

2. SDG Categories

SDG 3 – Good Health And Well-Being



3. Aim and Motivation

Aim

- To develop an Agentic AI system that converts scanned ECG images into digital signals, detects potential abnormalities, and generates an explainable ECG report.

Motivation

- Scanned ECGs are difficult to analyze directly using AI.
- ECG digitization enables automated signal analysis.
- Transformer models can detect abnormal ECG patterns.
- Agentic AI and RAG can provide explainable, evidence-based insights.

4. Research Questions

1. How effectively can a 1D ECG Transformer detect potential abnormal ECG patterns?

2. How can Agentic AI and Explainable AI improve the interpretability of ECG analysis?

3. How can RAG and confidence assessment support reliable, evidence-based ECG reporting?

5. Title Justification

The title **“Agentic AI-Based ECG Abnormality Detection Using 1D Transformer and Explainable Report Generation”** represents the core objective, computational approach, and outcome of the proposed project. **“Agentic AI”** refers to the use of AI agents to coordinate different stages of the system, including ECG quality assessment, knowledge retrieval, confidence evaluation, and report generation. **“ECG Abnormality Detection”** defines the primary objective of identifying potential abnormal patterns from ECG signals. **“1D Transformer”** represents the deep learning model used to analyze the digitized ECG waveform and capture important temporal patterns associated with abnormalities. **“Explainable Report Generation”** refers to converting the model’s predictions and relevant findings into an interpretable report that provides understandable insights into the detected abnormalities. Thus, the title clearly reflects the **AI-agent-based architecture, ECG abnormality detection task, 1D Transformer model, and explainable reporting component** of the proposed research.

6. Objectives

- To develop an image-processing framework for preprocessing scanned ECG images and extracting ECG waveforms as digital signals.
- To develop and evaluate a 1D ECG Transformer for identifying potential abnormal patterns from the digitized ECG signals.
- To integrate Agentic AI for ECG quality assessment, explainability, biomedical knowledge retrieval, and confidence assessment.
- To generate an explainable and evidence-based ECG report that can support clinical decision-making.

7. Scope

- The project is limited to detecting ECG abnormalities represented in the selected dataset.
- The system focuses on digital ECG signal analysis and does not cover all types of cardiac diseases or guarantee clinical diagnosis.
- The project focuses on AI-based ECG analysis and explainable report generation, excluding real-time hospital deployment and direct patient monitoring.

8. Timeline Chart

Table No. : Timeline chart for Mini Project work

S.No.	Activity	Jul	Aug	Sep	Oct	Nov	Dec
1	Broad domain and problem exploration	✓					
2	Literature survey and research paper analysis	✓	✓				
3	Problem statement finalization and requirement analysis		✓				
4	Dataset identification and feasibility study		✓				
5	Data collection and preprocessing		✓	✓			
6	Graph construction and representation learning						
7	Deep learning model development and training						
8	Error-aware retraining and false-positive reduction						
9	Performance evaluation and analysis						
10	Documentation, research paper and final presentation						

9. Introduction

9.1. Background

- Digital ECG signals contain important temporal patterns that can be analyzed for detecting potential cardiac abnormalities.
- A 1D ECG Transformer can capture temporal dependencies in ECG signals to identify potential abnormal patterns.
- Agentic AI, Explainable AI, and RAG can support ECG interpretation and generate explainable, evidence-based reports.

9.2. Problem Statement

- Manual analysis of ECG signals can be time-consuming and may make it difficult to consistently identify subtle abnormal patterns.
- An intelligent framework is required to automatically analyze digital ECG signals, identify potential abnormalities, assess prediction confidence, and provide interpretable results.
- The proposed framework uses a **1D ECG Transformer** for abnormality detection and an **Agentic AI-based approach with Explainable AI, RAG, and confidence assessment** to generate an explainable and evidence-based ECG report.

10.1 Study on Existing Technologies

Title: Automatic Digitization of Paper Electrocardiograms – A Systematic Review [1]
Journal Details: Journal of Electrocardiology, 2023, Elsevier.
Dataset: The paper is a systematic review of existing ECG digitization approaches; it discusses the lack of common databases and standardized evaluation metrics rather than proposing a single dataset.
Description: The paper reviews methods for converting paper or scanned ECG recordings into digital ECG signals. It discusses challenges such as skew correction, binarization, and signal extraction, along with recent deep-learning and automated approaches.
Advantages:
•Provides a comprehensive review of automatic ECG digitization techniques.
•Identifies important challenges in ECG image preprocessing and signal extraction.
•Highlights the potential of deep learning and automation for ECG digitization.
Gaps Addressed:
•Identifies the lack of standardized datasets and evaluation metrics for comparing ECG digitization methods.
•Highlights the limited availability of source code and software for existing approaches.
Limitations:
•Focuses mainly on ECG digitization rather than ECG abnormality detection.
•Does not address 1D Transformer-based ECG analysis, Agentic AI, or explainable report generation.
•Standardized datasets and evaluation methods for ECG digitization remain limited.

10.2 Study on Existing Technologies

Title: Digitizing Paper Electrocardiograms: Status and Challenges [2]
Journal Details: Journal of Electrocardiology, 2017, Elsevier.
Dataset: The paper is a literature review and does not use a specific dataset.It reviews existing ECG digitization algorithms, applications, criticisms, and standardization efforts.
Description: The paper presents a literature review of methods for converting paper ECG recordings into digital ECG signals. It compares different conversion algorithms and discusses their applications, key improvements, standardization efforts, and remaining challenges in ECG digitization.
Advantages:
•Provides a comprehensive review of existing ECG digitization methods.
•Discusses different conversion algorithms and their applications.
•Highlights important improvements and standardization efforts in ECG digitization.
Gaps Addressed:
•Identifies challenges and limitations in existing ECG conversion algorithms.
•Highlights the need for improved standardization of ECG digitization methods.
Limitations:
• Focuses mainly on ECG digitization rather than automated ECG abnormality detection.
• Does not address 1D Transformer-based ECG signal analysis.
•Does not cover Agentic AI, RAG, Explainable AI, or explainable ECG report generation.

10.3 Study on Existing Technologies

Title: An Explainable Deep Learning Framework for Trustworthy Arrhythmia Detection from ECG Signals [3]

Journal Details: Scientific Reports, 2025, Nature.

Dataset: MIT-BIH Arrhythmia Database (MITDB), PTB Diagnostic ECG Database (PTBDB), and Noise Stress Test Database (NSTDB).

Description: The paper presents an explainable deep-learning framework for arrhythmia detection from ECG signals. It uses **CNN and DNN architectures** along with signal preprocessing and data-balancing techniques such as ADASYN, SMOTE, SMOTETomek, and Random Over-Sampling. **SHAP, LIME, and Feature Importance Analysis** are used to explain the model's predictions and improve clinical interpretability.

Advantages:

- Achieves high arrhythmia classification performance on multiple benchmark ECG datasets.
- Uses multiple data-balancing techniques to improve model generalization.
- Integrates SHAP, LIME, and feature importance analysis for explainable predictions.

Gaps Addressed:

- Provides explainability for deep-learning-based ECG arrhythmia detection using SHAP, LIME, and feature importance.
- However, the framework primarily uses **CNN and DNN models** and does not explore **1D Transformer-based temporal modeling** for ECG abnormality detection.
- It does not integrate **Agentic AI, RAG-based biomedical knowledge retrieval, confidence assessment, and explainable report generation** into a unified ECG analysis workflow.

Limitations:

- Focuses on CNN and DNN architectures rather than Transformer-based ECG models.
- Explainability is mainly provided through individual XAI techniques rather than an agentic framework.
- Does not provide RAG-based evidence retrieval or automated evidence-based ECG report generation.

10.4 Study on Existing Technologies

Title: An Explainable Artificial Intelligence Framework for Trustworthy Arrhythmia Detection from Heart Rhythm Data [4]
Journal Details: Healthcare Analytics, 2026, Elsevier.
Dataset: Chapman ECG Dataset and Ningbo ECG Dataset.
Description: The paper proposes **EXGnet**, an explainable ECG arrhythmia classification framework for single-lead ECG signals. It uses a multiresolution block to capture short- and long-term signal features and incorporates XAI supervision during training to focus on clinically relevant ECG regions. Quantitative ECG features are also used during training while maintaining a lightweight model for inference and edge deployment.
Advantages:
•Provides accurate arrhythmia classification while incorporating explainability.
•Captures both short-term and long-term ECG signal features using a multiresolution architecture.
•Designed for lightweight and resource-efficient deployment on portable or edge devices.
Gaps Addressed:
•Provides XAI-guided ECG classification, but does not use a **1D Transformer architecture** to model long-range temporal dependencies in ECG signals.
•Focuses on explainable classification but does not integrate **Agentic AI for coordinating ECG quality assessment, confidence assessment, knowledge retrieval, and report generation**.
•Does not incorporate **RAG-based biomedical knowledge retrieval** to provide evidence-based explanations for detected abnormalities.
Limitations:
•The framework is designed specifically for single-lead ECG signals.
•Uses a specialized EXGnet architecture rather than a Transformer-based model.
•Does not provide an Agentic AI-based workflow for automated explainable ECG report generation.

10.5 Study on Existing Technologies

Title: BiomedRAG: A Retrieval Augmented Large Language Model for Biomedicine [5]
Journal Details: Journal of Biomedical Informatics, 2025, Elsevier.
Dataset: Four biomedical NLP tasks using eight biomedical datasets.
Description: The paper proposes **BiomedRAG**, a retrieval-augmented generation framework designed for biomedical applications. Instead of retrieving information only at the sentence or paragraph level, it retrieves relevant **chunk-based biomedical documents** and provides them to the LLM for generating more accurate responses. The framework improves the performance of biomedical NLP tasks and supports more accurate and adaptable biomedical LLM applications.
Advantages:
•Improves the accuracy of biomedical LLM applications through retrieval-augmented generation.
•Uses relevant chunk-based biomedical documents to reduce irrelevant information and retrieval noise.
•Does not integrate **Agentic AI, ECG-specific XAI, confidence assessment, or automated ECG report generation**.
Gaps Addressed:
•Demonstrates how RAG can improve the reliability and accuracy of LLMs in the biomedical domain.
•Provides a foundation for integrating biomedical knowledge retrieval into the proposed ECG reporting system.
•Can support evidence-based explanations of detected ECG abnormalities when combined with an ECG analysis model.
Limitations:
•Focuses on general biomedical NLP tasks rather than ECG abnormality detection.
•Does not use ECG signals or a **1D ECG Transformer** for abnormality classification.
•Does not integrate **Agentic AI, ECG-specific XAI, confidence assessment, or automated ECG report generation**.

10.6 Study on Existing Technologies

Title: ECG-FM: An Open Electrocardiogram Foundation Model [6]
Journal Details: JAMIA Open, Volume 8, Issue 5, October 2025, Oxford University Press.
Dataset: 1.5 million 12-lead ECGs for pretraining; MIMIC-IV-ECG for downstream evaluation and benchmarking.
Description: The paper introduces ECG-FM, an open-weight, Transformer-based foundation model pretrained using masked reconstruction and contrastive learning. It evaluates ECG interpretation and reduced left ventricular ejection fraction prediction. The model demonstrates strong label efficiency and cross-dataset generalization, achieving an AUROC of 0.996 for atrial fibrillation detection.
Advantages:
•Uses Transformer-based learning to capture meaningful ECG signal representations.
•Demonstrates strong AFib detection performance and generalization across datasets.
•Provides publicly available model weights, code, and benchmarks for reproducible research.
Gaps Addressed:
•Demonstrates the effectiveness of pretrained Transformer representations for ECG analysis, providing a foundation for 1D ECG abnormality detection.
•Highlights the potential to reduce dependence on large labelled datasets through pretraining and fine-tuning.
Limitations:
•Focuses on ECG interpretation and prediction rather than Agentic AI-coordinated ECG analysis and report generation.
•Does not describe the complete integration of biomedical RAG, confidence assessment, and automated evidence-based explanations proposed in our project.

10.7 Study on Existing Technologies

Title: Bridging Clinical Knowledge and AI: An Interpretable Transformer Framework for ECG Diagnosis [7]
Journal Details: npj Digital Medicine, 2025, Nature Portfolio.
Dataset: MIT-BIH Arrhythmia Database (8 classes) and Chapman ECG Dataset (12 classes).
Description: The paper proposes MDOT (Momentum Distillation Oscillographic Transformer), which combines physician-derived clinical knowledge with Transformer-based ECG analysis. It incorporates clinical features such as heart rate, QRS duration, ST-segment changes, and QTc interval. A novel oscillographic module converts 1D ECG signals into 2D representations, while attention-based heatmaps highlight diagnostically relevant regions. The model achieves accuracies of 99.53% on MIT-BIH and 99.03% on Chapman under strict inter-patient evaluation.
Advantages:
•Integrates clinical knowledge into Transformer-based ECG classification.
•Uses attention-based heatmaps to improve interpretability.
•Demonstrates high classification accuracy and considers lightweight edge deployment.
Gaps Addressed:
•Demonstrates how clinical features and interpretability can improve Transformer-based ECG diagnosis.
•Provides a foundation for developing explainable ECG abnormality detection systems.
Limitations:
•Converts 1D ECG signals into 2D oscillographic representations rather than relying solely on direct 1D Transformer processing.
•Focuses on ECG classification and interpretability rather than Agentic AI-coordinated analysis, biomedical RAG, confidence assessment, and automated evidence-based report generation.

10.8 Study on Existing Technologies

Title: A Joint CNN-Bi-LSTM-Transformer Architecture with SHAP Explanations for Multi-Label Arrhythmia Detection from 12-Lead ECGs [8]
Journal Details: Scientific Reports, 2026, Nature Portfolio.
Dataset: PhysioNet/Computing in Cardiology Challenge 2020 dataset, containing more than 43,000 ECG recordings across 27 arrhythmia classes.
Description: The paper proposes a hybrid deep-learning architecture combining CNN, Bi-LSTM, and Transformer layers to extract morphological, temporal, and spatial patterns from 12-lead ECG signals. It integrates SHAP to explain predictions by identifying relevant waveform segments and ECG leads. The model reports 91.24% accuracy, a macro-F1 score of 83.01%, and AUC exceeding 0.98 for certain arrhythmias, including atrial fibrillation.
Advantages:
•Combines CNN, Bi-LSTM, and Transformer layers to learn different ECG signal characteristics.
•Uses SHAP to improve the interpretability of arrhythmia predictions.
•Addresses class imbalance and reports low model footprint and inference latency.
Gaps Addressed:
•Demonstrates the integration of hybrid deep learning and SHAP-based explanations for ECG arrhythmia detection.
•Provides a comparison for evaluating the proposed 1D ECG Transformer against a more complex hybrid architecture.
Limitations:
•Uses a hybrid CNN-Bi-LSTM-Transformer architecture rather than a standalone 1D ECG Transformer.
•Focuses on arrhythmia classification and SHAP explanations without describing the complete integration of Agentic AI, biomedical RAG, confidence assessment, and automated evidence-based report generation.

10.7 Study on Existing Technologies

S. No.	Article Title	Journal Details	Algorithms/ Models	Dataset	Advantage	Disadvantage
1	Automatic Digitization of Paper Electrocardiograms – A Systematic Review	Journal of Electrocardiology, 2023, Elsevier	ECG digitization, image preprocessing, deep-learning-based extraction methods	Systematic review; no single dataset	Reviews ECG digitization methods. Identifies preprocessing and signal extraction challenges.	Focuses on digitization rather than abnormality detection. Limited standardization and code availability.
2	Digitizing Paper Electrocardiograms: Status and Challenges	Journal of Electrocardiology, 2017, Elsevier	Existing ECG digitization and conversion algorithms	Literature review; no specific dataset	Compares conversion methods. Highlights standardization challenges and potential improvements.	Does not focus on automated ECG abnormality detection or Transformer-based analysis.

10.8 Study on Existing Technologies

S. No.	Article Title	Journal Details	Algorithms/ Models	Dataset	Advantage	Disadvantage
3	An Explainable Deep Learning Framework for Trustworthy Arrhythmia Detection from ECG Signals	Scientific Reports, 2025, Nature	CNN, DNN, SHAP, LIME, Feature Importance Analysis; SMOTE and other balancing methods	MIT-BIH, PTBDB, NSTDB	Supports explainable arrhythmia classification. Uses data balancing and multiple XAI techniques.	Does not integrate 1D Transformer, Agentic AI, RAG, or automated report generation.
4	An Explainable Artificial Intelligence Framework for Trustworthy Arrhythmia Detection from Heart Rhythm Data	Healthcare Analytics, 2026, Elsevier	EXGnet, multiresolution block, XAI supervision	Chapman ECG Dataset, Ningbo ECG Dataset	Captures short- and long-term ECG features. Supports interpretable and lightweight classification.	Focuses on single-lead classification and does not include the proposed Agentic AI + RAG reporting workflow.

10.9 Study on Existing Technologies

S. No.	Article Title	Journal Details	Algorithms/ Models	Dataset	Advantage	Disadvantage
5	BiomedRAG: A Retrieval Augmented Large Language Model for Biomedicine	Journal of Biomedical Informatics, 2025, Elsevier	Retrieval-Augmented Generation (RAG), large language models (LLMs), chunk-based document retrieval	Eight datasets across four biomedical NLP tasks	Improves biomedical LLM responses using retrieved knowledge. Supports evidence-grounded generation.	Does not perform ECG signal classification or provide an ECG-specific Agentic AI reporting system.
6	ECG-FM: An Open Electrocardiogram Foundation Model	JAMIA Open, 2025, Oxford University Press	Transformer-based foundation model, masked reconstruction, contrastive learning	1.5 million 12-lead ECGs; MIMIC-IV-ECG benchmark	Supports ECG interpretation, label-efficient learning, and cross-dataset generalization.	Does not describe the complete proposed Agentic AI, RAG, and explainable report-generation workflow.

10.9 Study on Existing Technologies

S. No.	Article Title	Journal Details	Algorithms/ Models	Dataset	Advantage	Disadvantage
7	Bridging Clinical Knowledge and AI: An Interpretable Transformer Framework for ECG Diagnosis	npj Digital Medicine, 2025, Nature Portfolio	MDOT, Transformer, momentum distillation, oscillographic representation, attention-based heatmaps	MIT-BIH Arrhythmia Database, Chapman ECG Dataset	Integrates clinical features into Transformer-based ECG diagnosis. Provides interpretable ECG regions.	Converts 1D signals into 2D representations and does not include the complete proposed Agentic AI + RAG workflow.
8	A Joint CNN-Bi-LSTM-Transformer Architecture with SHAP Explanations for Multi-Label Arrhythmia Detection from 12-Lead ECGs	Scientific Reports, 2026, Nature Portfolio	CNN, Bi-LSTM, Transformer, SHAP	PhysioNet/Computing in Cardiology Challenge 2020; over 43,000 recordings and 27 arrhythmia classes	Combines morphological and temporal feature learning with SHAP-based explanations. Reports low inference latency.	Uses a hybrid architecture rather than a standalone 1D Transformer and does not integrate Agentic AI, RAG, or automated evidence-based reporting.

9. Epics Outcome

Research paper titled “Emergency Communication System for Hikers Using LoRa” was published in the 2025 1st International Conference on Advancement in Futuristic Technologies (ICAFT 2025). The paper was published in IEEE Conference Proceedings.

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